Sem IV	OE	Open Elective 1 (From Other School) Open-Source Technologies Introduction to Data Science	3	0	0	3	3
	PC	Design Thinking	0	0	2	2	1
	PC	Probability and Statistics Theory	3	0	0	3	3
	PC	Computer Organization and Architecture	3	1	0	4	4
	PC	Software Engineering	3	0	0	3	3
	PC	Software Engineering Laboratory	0	0	2	2	1
	PC	Theory of Computation	3	1	0	4	4
	PC	Design and Analysis of Algorithm	3	0	0	3	3
	PC	Design and Analysis of Algorithm Laboratory	0	0	2	2	1
			18	**2**	**6**	**24**	**23**
Sem V	OE	Open Elective 2 ((NPTEL/SWAYAM/MOOC))	3	0	0	3	3
	ESC	Introduction to Artificial Intelligence (Department Specific)	3	0	0	3	3
	HSC	Engineering Economics	3	0	0	3	3
	PE	Program Elective 1	3	0	0	3	3
	PC	Computer Networks	3	0	0	3	3
	PC	Computer Networks Laboratory	0	0	2	2	1
	PC	Compiler Design	3	0	0	3	3
	PC	Compiler Design Laboratory	0	0	2	2	1
	PC	Operating System	3	0	0	3	3
	PC	Operating System Laboratory	0	0	2	2	1
			21	**0**	**6**	**27**	**24**
Sem VI	OE	Open Elective 3 (From Other Department of FoET) Web Development Object Oriented Programming & Design	3	0	0	3	3
	PE	Program Elective 2	3	0	0	3	3
	PE	Program Elective 3	3	0	0	3	3
	PC	Cryptography and Network Security	3	0	0	3	3
	PC	Cryptography and Network Security Laboratory	0	0	2	2	1
	PC	Web & Mobile Development Essentials	1	0	0	1	1
	PC	Web & Mobile Development Essentials Laboratory	0	0	4	4	2
	PC	Distributed Computing	3	0	0	3	3
	PC	Distributed Computing Laboratory	0	0	2	2	1
			16	**0**	**8**	**24**	**20**
Sem VII	Pro	Summer Internship	0	0	0	0	2
	OE	Open Elective 4 (From Other Department of FoET) Green IT	3	0	0	3	3

		Global Data Security & Privacy Laws					
	ESC	Industry 4.0 (Department Specific)	2	0	0	2	2
	ESC	Industry 4.0 Laboratory (Department Specific)	0	0	2	2	1
	PE	Program Elective 4	3	0	0	3	3
	PE	Program Elective 5	3	0	0	3	3
	PC	Machine Learning	3	0	0	3	3
	PC	Machine Learning Laboratory	0	0	2	2	1
	PC	Cyber Laws and Ethics	1	0	0	1	1
	Pro	Seminar					1
			15	**0**	**4**	**19**	**20**
Sem VIII		**Course Name**	**Th**	**Tut**	**Pra**	**Hrs**	**Cr**
	Pro	Major/Comprehensive Project					**12**

Total Credits: 160

List of Electives (Track-based)

	Artificial Intelligence/ Machine Learning Track	**Industry Track**	**Speech, Vision & Text Track**	**Computing Track**	**Communication Track**
PE-1	Data Mining and Data Warehousing	Object Oriented Modelling and Design	Computer Graphics	Advanced Data Structure and Algorithms	Data Communication
PE-2	Soft Computing	UI/UX Design	Digital Image Processing	Blockchain Technology	Mobile Computing
PE-3	Deep Learning	Secure Software Engineering	Computer Vision	Big Data Analytics	Wireless Sensor Networks
PE-4	Agent based Learning	Agile and DevOps	Natural Language Processing	Cloud Computing	Internet of Things
PE-5	Machine Learning in Cyber Security	Web Application Testing	Speech Processing	Quantum Computing	Autonomous Systems

7th Semester

24CS422T					Global Data Security and Privacy Laws					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	-	-	100

COURSE OBJECTIVES

- To understand the need of Global Data Security and Privacy Laws.
- To be aware about global data protection regulations and compliance framework.
- To be able to implement data security measures.
- To be aware about the impact of emerging technologies on data privacy and security.

UNIT 1: Introduction to Global Data Security and Privacy Laws Overview of data security and privacy concepts, Introduction to major global data protection regulations (e.g., GDPR, CCPA, LGPD), Understanding the importance of data protection in a global context, Case studies highlighting the impact of data breaches on businesses and individuals	**10 Hrs.**
UNIT 2: Compliance Frameworks and Regulations Key global data protection regulations and frameworks, Comparative analysis of GDPR, CCPA, LGPD, and other relevant laws, Understanding the requirements for data collection, processing, and storage, Practical guidance on achieving compliance with international data security standards.	**11 Hrs.**
UNIT 3: Implementing Data Security Measures Strategies for implementing data security measures in compliance with global regulations, Encryption techniques and best practices for data protection, Data breach response and incident management procedures, Role of cybersecurity technologies in safeguarding sensitive data.	**11 Hrs.**
UNIT 4: Emerging Trends and Future Challenges Exploration of emerging trends in global data security and privacy laws, Impact of emerging technologies (e.g., AI, IoT) on data protection, Addressing challenges posed by cross-border data transfers, Future directions in global data security and privacy regulations and their implications for businesses	**10 Hrs.**
	42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand the fundamental concepts of data security and privacy.
CO2 - Interpret the requirements and obligations imposed by GDPR, CCPA, and LGPD on organizations.
CO3 - Apply the principles of data protection to real-world scenarios and case studies.
CO4 - Assess the impact of data breaches on organizations and individuals, and formulate appropriate response strategies.
CO5 - Develop data protection policies and procedures tailored to the requirements of global data security laws.
CO6 - Critically evaluate emerging trends in global data security and privacy laws.

TEXT/REFERENCE BOOKS

1. Paul Voigt and Axel von dem Bussche, "Understanding the GDPR: A Practical Guide to Global Privacy and Data Security Law", published by Wolters Kluwer.
2. Lisa J. Sotto (Editor), "Twentieth Annual Data Privacy and Security Law", published by Practicing Law Institute New York.
3. Peter Carey, "Data Protection: A Practical Guide to UK and EU Law", published by Oxford University Press.
4. Jeff Kosseff, "Cybersecurity Law", published by Wiley.

24CS421T					Green IT					
Teaching Scheme					**Examination Scheme**					
L	**T**	**P**	**C**	**Hrs/Week**	**Theory**			**Practical**		**Total Marks**
					MS	**ES**	**IA**	**LW**	**LE/Viva**	
3	**0**	**0**	**3**	**3**	**25**	**50**	**25**	**--**	**--**	**100**

COURSE OBJECTIVES

- Students should gain knowledge of the environmental impact of traditional IT infrastructure, including energy consumption, carbon emissions, e-waste generation, and resource depletion.
- Learn about sustainability principles and how they apply to the design, deployment, and management of IT systems and services.
- Understand relevant environmental policies, regulations, and standards related to IT sustainability, both at the national and international levels.
- Develop practical skills for implementing Green IT strategies, including energy auditing, environmental impact assessment, and sustainability reporting.

UNIT 1 INTRODUCTION TO GREEN COMPUTING Introduction, Environmental Concerns and Sustainable Development, Environmental Impacts of IT, Holistic Approach to Greening IT. Green Devices and Hardware with Green Software.	**11 Hrs.**
UNIT 2 GREEN ENTERPRISE Green Enterprises and the Role of IT - Introduction, Organization and Enterprise Greening, Information systems in Greening.	**11 Hrs.**
UNIT 3 MANAGING AND REGULATING GREEN IT Strategizing Green Initiatives, Implementation of Green IT, Communication and Social media. Regulating the Green IT: Laws, Standards and Protocols .	**11 Hrs.**
UNIT 4 GREEN IT CASE STUDIES Awareness to implementations, Research and Development directions. Worldwide Green IT Case Studies	**9 Hrs.**
	42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Define Green IT with its different dimensions and Strategies.
CO2 - Classify Green devices and hardware along with its green software methodologies.
CO3 - Apply the various green enterprise activities, functions and their role with IT.
CO4 - Analyze the concepts of how to manage the green IT with necessary components.
CO5 - Select the various laws, standards and protocols for regulating green IT.
CO6 - Discuss the various key sustainability and green IT trends.

TEXT/REFERENCE BOOKS

1. Toby J. Velete, Anthony T. Velete, Robert Elsenpeter, “Green IT – Reduce Your Information System’s Environmental Impact While Adding to the Bottom Line”, McGraw-Hill
2. John Lamb, “The Greening of IT – How Companies Can Make a Difference for the Environment”, IBM Press
3. San Murugesan, G.R. Gangadharan, “Harnessing Green IT Principles and Practices”, Wiley Publication
4. Richard Maltzman and David Shirley, “Green Project Management”, CRC Press

24CS401T					Industry 4.0					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
2	0	0	2	2	50	50	25	--	--	100

COURSE OBJECTIVES

- To impart basic ideas in Industry 4.0.
- To provide students with knowledge of designing Industrial 4.0 Systems for various applications.
- Learn the techniques and tools to offer solutions of Industry 4.0 applications.
- Explore the design and analysis of Industry 4.0 systems tailored for different verticals, enabling students to develop innovative solutions for sustainable and efficient operations.

UNIT I: Introduction and Basic concepts of Industry 4.0 Introduction to Industry 4.0, Definition, General framework, Application areas, Introduction to the Evolution of Industrial revolutions, Key features, Need and benefits, Introduction to Industry 4.0 core technologies: Big data, Advanced Robotics, Simulation, Integration, Internet of Things (IoT), Artificial Intelligence (AI), Cybersecurity, Cloud computing, Additive manufacturing and Augmented Reality. Transformation of industrial processes through the integration of modern technologies such as sensing and actuation, communication, and computational processing.	**6 Hrs.**
UNIT 2: Enabling Technologies for Industry 4.0 Introduction to Enabling Technologies for Industry 4.0, Technology Road Map, Introduction to Connectivity (wired and wireless), Introduction to Networking, Wireless Sensor Networks, Embedded Systems for IIoT (Industrial IoT), Smart and Connected Business Perspective.	**7 Hrs.**
UNIT 3: Introduction to Advanced Technologies and Emerging Trends in Industry 4.0 The Internet of Things (IoT), Characteristics, Physical design of IoT, Logical Design of IoT, Challenges for IoT, Sensors and Actuators for IoT Applications, Introduction to Digital Twin, Introduction to Cyber Physical Systems (Robotics), Cloud Computing, Visualization, Basics of Data Analytics, Artificial intelligence, Concept of Drones/Unmanned Aerial Vehicles (UAVs), Cyber security for Industry 4.0.	**8 Hrs.**
UNIT 4: Case Studies and Application for Industry 4.0 Various application of Industrial IoT: Smart City, Smart Grid, Smart Transportation, Smart Manufacturing, Smart Healthcare	**7 Hrs.**
	28 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 – Understand the core elements and basic technologies for Industry 4.0.
CO2 – Understand the traditional business approach by integrating the data and intelligence.
CO3 – Understand the traditional industries with intelligent and automated machines.
CO4 – Utilize data and intelligence for the development of Smart World.
CO5 – Apply the different computational techniques and algorithms for realizing Industry 4.0.
CO6 – Apply the concept, significance and means to achieve sustainable development.

TEXT/REFERENCE BOOKS

1. The Concept Industry 4.0: An Empirical Analysis of Technologies and Applications in Production Logistics by Christoph Jan Bartodziej
2. Ustundag Alp, and Emre Cevikcan, Industry 4.0: Managing the Digital Transformation, Springer
3. Industry 4.0: Entrepreneurship and Structural Change in the New Digital Landscape, By Springer
4. Bali, V., Banerjee, K., Kumar, N., Gour, S., & Chawla, S. K. (Eds.). Industry 4.0, AI, and Data Science: Research Trends and Challenges.

24CS401P					Industry 4.0 Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/ Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

PREREQUISITES: Python Programming

COURSE OBJECTIVES

- Gain a basic understanding of Industry 4.0 principles and technologies.
- Learn to design and implement simple IoT systems for everyday applications.
- Explore introductory concepts in advanced technologies like cloud computing and AI.
- Discover practical examples of IoT applications in smart cities, transportation, healthcare, and manufacturing for beginners.

LAB EXPERIMENTS

1. **Introduction to programming -** Basic computations using Python programming
2. **Introduction to interfacing -** Introduction to Interfacing of Basic I/O devices
3. **Internet of Thing Study -** Designing the Publish Subscribe system using MQTT and Paho on Mosquito Broker
4. **Understanding of Interface and Controller-** Demo of Interfacing and controlling LED using MQTT protocol and HIVEMQ broker
5. **Communication protocol –** Demo of Interfacing Bluetooth/ GSM module with Arduino platform
6. **Machine Learning in Industry 4.0 –** Implementing a Basic Machine Learning Model on Arduino
7. **Cloud Computing in Industry 4.0 –** Study of Containerising a python application using docker
8. **Learning of UAVs/Drones –** UAVs application programming using python
9. **Development of application for Industry 4.0 -** Demo of line follower autonomous vehicle.
10. **Collaborative Learning –** Case studies/Project based on topics in Unit 4.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Recall key concepts and principles related to IoT, networking, embedded systems, and data analytics.

CO2 - Demonstrate comprehension of the fundamental technologies and methodologies used in Industry 4.0 applications.

CO3 - Apply theoretical knowledge to design and implement IoT systems and experiments in the laboratory setting.

CO4 - Analyze data collected from IoT experiments and draw conclusions based on observed results.

CO5 - Design and develop projects using industry 4.0 technologies to address real-world challenges.

CO6 - Evaluate the effectiveness and efficiency of industry 4.0 solutions developed during the lab sessions, considering factors such as performance, scalability, and security.

TEXT/REFERENCE BOOKS

1. Moshayedi, Ata Jahangir, Amin Kolahdooz, and Liefa Liao. Unity in Embedded System Design and Robotics: A Step-by-step Guide. Chapman and Hall/CRC
2. Kulkarni, Shrirang Ambaji, Varadraj P. Gurupur, and Steven L. Fernandes. Introduction to IoT with Machine Learning and Image Processing Using Raspberry Pi. Chapman and Hall/CRC
3. Practical Python Programming for IoT by Gary Smart
4. Programming Python, 4th Edition by Mark Lutzs

24CS402T					Machine Learning					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	-	-	100

COURSE OBJECTIVES

- To equip students with fundamental machine learning concepts.
- To develop skills of using recent machine learning techniques for solving practical problems
- Understand the methods for evaluating model performance.

UNIT 1: Essential Of Machine Learning Linear Algebra: Vectors, Matrices, Linear Transformations, Eigenvalue and Eigenvectors, Calculus: Derivatives, Gradients and Optimization Techniques Like Gradient Descent, Probability and Statistics: Probability Distributions, Expected Values, Variance, Hypothesis Testing and Bayesian Inference, Multivariate Calculus: Partial Derivatives and Gradients of Multivariable Functions.	**12 Hrs.**
UNIT 2: Supervised Machine Learning History of Machine Learning, Regression & Its Type, Gradient Descent & Its Type, Overfitting and Underfitting, Regularization: Lasso, Ridge, Elastic Net, Kernel Ridge Regression, Support Vector Regression, Generalization, Classification Problems, Decision Boundaries, Decision Tree, Nearest Neighbor Methods, Bayes Optimal Decisions, Naive Bayes Classifier, Logistic Regression, Support Vector Machines (SVM), Support Vector Machines and Large-Margin Classifiers, Supervised Model Evaluation Metrics, Artificial Neural Networks, Imbalanced data handling techniques: SMOTE, ADASYN.	**12 Hrs.**
UNIT 3: Unsupervised Machine Learning K-Mean Clustering, Density-Based Clustering: DBSCAN, OPTICS, Hierarchical Clustering with Advanced Linkage Methods, Spectral Clustering for Non-Convex Clusters, Dimensionality Reduction-PCA, Principal Components Analysis (PCA), EM Algorithm, Gaussian Mixture Models, Unsupervised Model Evaluation Metrics.	**08 Hrs.**
UNIT 4: Semi-Supervised Learning And Ensemble Methods Role of Semi-Supervised Learning, Self-training, Co-Training and Multi-View Learning Approaches, Label Propagation and Graph-Based Methods for Semi-Supervised Learning, Rationale for Ensemble Method, Methods for Constructing an Ensemble Classifier, Bagging, Boosting, Random Forests, Empirical Comparison Among Ensemble Methods.	**10 Hrs.**
	42 Hrs.

COURSE OUTCOMES

On completion this course, students will be able to:

CO - 1 Explain the stages of the machine learning pipeline and classify machine learning into supervised, unsupervised, and semi-supervised learning.

CO - 2 Apply feature engineering techniques to preprocess and manipulate data for machine learning tasks.

CO - 3 Evaluate the performance of the different Machine Learning approaches

CO - 4 Identify and mitigate overfitting and underfitting issues in supervised learning models using techniques like regularization.

CO - 5 Understand the principles of unsupervised learning and apply them to solve real-world problems

CO - 6 Justify the use of ensemble methods and explain the process of constructing ensemble classifiers

TEXT/REFERENCE BOOKS

1. Tom M. Mitchell, “Machine Learning”, McGraw-Hill Education.
2. Christopher M. Bishop, "Pattern Recognition and Machine Learning", by Springer.
3. Amanda Casari, Alice Zheng, “Feature Engineering for Machine Learning”, O’Reilly.

24CS402P					Machine Learning Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

4. Andreas Muller, "Introduction to Machine Learning with Python: A Guide for Data Scientists", Shroff/O'Reilly.
5. M. Gopal "Applied Machine Learning", McGraw-Hill

PREREQUISITES:

- Familiarity with programming languages such as Python.

COURSE OBJECTIVES

- To familiarize students with popular tools and libraries used in applied machine learning.
- To develop problem-solving skills and critical thinking abilities to approach and tackle different types of machine learning problems effectively.
- How to evaluate the performance of machine learning models using appropriate metrics and validation techniques

LIST OF EXPERIMENTS

Data sets can be taken from standard repositories (https://archive.ics.uci.edu/ml/datasets.html) or constructed by the students. Preferred Programming Language & Platform: Python/R, TensorFlow.

1. Implement regression algorithms like and understand their performance metrics. Implement linear regression using gradient descent. Train the regression models on sample datasets and evaluate their performance using metrics like Mean Squared Error (MSE) and R-squared. Also explore different regularization techniques (e.g., Lasso, Ridge) and analyze their impact on model performance.
2. Implement classifiers such as Naive Bayes, logistic regression, decision trees, and SVMs. Train the classifiers on sample datasets and evaluate their performance using confusion matrices, accuracy, precision, recall, and F1-score. Visualize decision boundaries and analyze the model's behavior.
3. Apply imbalanced data handling techniques (SMOTE, ADASYN) on a dataset with class imbalance.
4. Implement k-means, hierarchical clustering, DBSCAN clustering etc.
5. Apply PCA for dimensionality reduction and visualize the reduced-dimensional data take suitable dataset to demonstrate the concept.
6. Implement Gaussian Mixture Models (GMM) for density estimation and clustering.
7. Implement ensemble methods such as bagging, boosting, and random forests. Train ensemble classifiers on sample datasets and compare their performance against individual classifiers. Conduct an empirical comparison among ensemble methods and analyze their strengths and weaknesses.
8. Apply machine learning techniques to real-world datasets and problems. Choose a real-world dataset relevant to the students' domain or interests. Perform exploratory data analysis (EDA) and feature engineering. Select appropriate machine learning models and train/test them on the dataset. Evaluate model performance and iterate on the model selection and tuning process.
9. Case study on the application of machine learning in various domains like NLP, Computer Vision, Speech and Audio.
10. Course project:Students are required to submit a course project that involves the development of a Machine learning system to provide a solution to a real-world problem using sample or real-time data sets.

COURSE OUTCOMES

On completion of this course, students will be able to:

CO - 1 Utilize machine learning tools and libraries such as scikit-learn and TensorFlow to develop machine learning models

CO - 2 Demonstrates proficiency in importing and exploring sample datasets using scikit-learn, and performs essential data preprocessing tasks

CO - 3 Evaluate a range of machine learning algorithms along with their implementation

CO - 4 Formulate machine learning problems corresponding to different applications.

CO - 5 Apply appropriate machine learning techniques to solve problems of moderate complexity

CO - 6 Develop machine learning-based solutions to real-world problems, optimize the models learned and report on the expected accuracy that can be achieved by applying the models

TEXT/REFERENCE BOOKS

1. Jeff Prosise " Applied Machine Learning and AI for Engineers", O'Reilly Media, Inc.

2. Sebastian Raschka, Vahid Mirjalili, “Python Machine Learning”, Packt publication
3. Manohar Swamynathan, “Mastering Machine Learning with Python in Six Steps: A Practical Implementation Guide to Predictive Data Analytics Using Python”, Apress.

24CS403T					Cyber Laws and Ethics					
Teaching Scheme					**Examination Scheme**					
L	**T**	**P**	**C**	**Hrs/Week**	**Theory**			**Practical**		**Total Marks**
					MS	**ES**	**IA**	**LW**	**LE/Viva**	
3	**0**	**0**	**3**	**3**	**25**	**50**	**25**	**-**	**-**	**100**

COURSE OBJECTIVES

- To understanding the difference between physical crime prosecution and cybercrime prosecution.
- To understand the legal aspects of cyber activities and be aware about different cyber laws.
- To be aware about ethical and professional conducts for upcoming digital working atmosphere.
- To understand the basics of digital evidence handling and digital forensics.

UNIT 1: Introduction to Cyber Law and Ethics Overview of Cyber Law: Historical development and key principles, Legal Frameworks: International, national, and regional laws governing cyberspace, Ethical Dimensions: Understanding ethical theories and their application in the digital realm, Case Studies: Analysis of landmark cyber law cases and ethical dilemmas	**10 Hrs.**
UNIT 2: Legal Aspects of Cyber Activities Jurisdiction in Cyberspace: Challenges and solutions, Intellectual Property Rights: Copyright, trademark, and patent laws in the digital age, Cybercrimes and Prosecution: Legal responses to hacking, cyberterrorism, and online fraud, Privacy and Data Protection: Legal frameworks for safeguarding personal information online, IT ACT 2000 and its amendments.	**11 Hrs.**
UNIT 3: Ethical Considerations in Cyber Space Ethical Decision Making: Models and frameworks for ethical reasoning, Digital Citizenship: Responsibilities and rights of individuals in online communities, Cybersecurity Ethics: Balancing security measures with individual privacy rights, Emerging Ethical Issues: AI ethics and tools , algorithmic biases, and the ethics of autonomous systems	**11 Hrs.**
UNIT 4: Digital Forensics Principles and methodologies, Digital Evidence Collection: Techniques for preserving, collecting, and analyzing digital evidence, Forensic Tools and Techniques: Overview of software and hardware used in digital investigations, Case Studies in Digital Forensics: Real-world examples of digital forensic investigations and their legal implications.	**10 Hrs.**
	42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand the basic law terminology and need of international governance in case of cybercrime.
CO2 - Distinguish between normal cyber behaviour and cybercrime.
CO3 - Understand IT-ACT 2000 and its amendments.
CO4 - Applying ethical conducts in professional life and effectively preserving and exploring intellectual property.
CO5 - Analyse digital forensic cases through to explore real world digital forensic tools and digital evidence handling.
CO6 - Analyse use of cyber laws and ethical code of conduct methods with case studies.

TEXT/REFERENCE BOOKS

1. Vakul Sharma, “Information Technology Law and Practice: Law & Emerging Technology Cyber Law”, Universal Law Publishing.
2. David S. Wall, “Cybercrime: The Transformation of Crime in the Information Age”, Wiley Computer.
3. Anirudh Rastogi, “Cyber Law of Information Technology and Internet”, LexisNexis.
4. Eoghan Casey, “Handbook of Digital Forensics and Investigation” , Elsevier
5. ACM Code of Ethics and Professional Conduct
6. ACM/IEEE Software Engineering Code of Ethics and Professional Practice.

Program Elective Course Track 1

Artificial Intelligence/ Machine Learning Track

24CS431T					Agent-based Learning					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

PREREQUISITE: Machine Learning, Probability and Statistics, Linear Algebra, Deep Learning

COURSE OBJECTIVES

- Comprehensive understanding of fundamental concepts in reinforcement learning, including the agent-environment interaction.
- Develop skills to evaluate and analyze the performance of reinforcement learning models using appropriate metrics such as cumulative reward, convergence rate, and learning stability.
- To gain proficiency in designing and implementing reinforcement learning systems, including defining state and action spaces, designing reward functions, selecting appropriate algorithms, and tuning hyperparameters.
- To demonstrate the ability to formulate problems, design solutions, and interpret results in these domains.

UNIT 1 Introduction to Reinforcement Learning Overview of reinforcement learning (RL) concepts, terminology, and applications. History and evolution of RL, from early research to modern deep reinforcement learning. Basic components of RL: agent, environment, state, action, reward, and policy. Types of RL problems: episodic vs. continuing tasks, finite vs. infinite horizon, deterministic vs. stochastic environments. Exploration vs. exploitation trade-off and its importance in RL. Key algorithms: Multi-Armed Bandits, Markov Decision Processes (MDPs), and Dynamic Programming.	**10 Hrs.**
UNIT 2 Value-Based Methods Introduction to value-based methods for RL, focusing on estimating value functions. Temporal Difference (TD) learning and its connection to dynamic programming. Q-learning: the off-policy TD control algorithm. Deep Q-Networks (DQN) and its extensions for handling large state spaces. Double Q-learning and Dueling DQN architectures for improved stability and efficiency. Experience replay and target networks for stabilizing training.	**10 Hrs.**
UNIT 3 Policy-Based Methods Introduction to policy-based methods for RL, focusing on learning a policy directly. Policy Gradient methods: REINFORCE, Actor-Critic architectures. Advantage Actor-Critic (A2C) and Proximal Policy Optimization (PPO) algorithms. Trust Region Policy Optimization (TRPO) and its successor, Trust Region Policy Optimization (TRPO). Importance of exploration strategies in policy-based methods: entropy regularization and exploration bonuses. Applications and extensions of policy-based methods, such as natural gradient methods and distributional RL	**13 Hrs.**
UNIT 4 Model-Based and Advanced Topics Model-based RL: learning a model of the environment and using it for planning and decision making. Model-based RL algorithms: Model Predictive Control (MPC), Model-Based Policy Optimization (MBPO). Model-based versus model-free methods: trade-offs and applications. Exploration in model-based RL: uncertainty estimation and exploration strategies. **Advanced topics in RL:** meta-learning, hierarchical RL, imitation learning, multi-agent RL. Emerging trends and future directions in RL research and applications.	**9 Hrs.**
	42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand mathematical concepts and fundamental concepts in reinforcement learning, including the agent-environment interaction, Markov decision processes (MDPs), reward signals, and exploration-exploitation trade-offs.

CO2 - Apply a variety of reinforcement learning algorithms to solve sequential decision-making problems, including value iteration, policy iteration, Q-learning, deep Q-networks (DQN), policy gradient methods, and actor-critic architectures.

CO3 - Apply policy gradient methods such as REINFORCE and Actor-Critic architectures to learn policies in RL tasks.

CO4 - Analyze the performance of reinforcement learning models using appropriate metrics such as cumulative reward, convergence rate, and learning stability

CO5 - Evaluate the trade-offs between model-based and model-free reinforcement learning methods, considering factors such as sample efficiency, computational complexity, and generalization capabilities.

CO6 - Formulate reinforcement learning techniques to solve real-world problems in domains such as robotics, game playing, finance, healthcare, and autonomous systems.

TEXT/REFERENCE BOOKS

1. R. S. Sutton and A. G. Barto. Reinforcement Learning - An Introduction. MIT Press.
2. Maxim Lapan. Deep Reinforcement Learning Hands-On: Apply modern RL methods, with deep Q-networks, value iteration, policy gradients, TRPO, AlphaGo Zero and more. Packt Publishing.
3. Enes Bilgin. Mastering Reinforcement Learning with Python, Packt Publishing.

24CS436T					Machine Learning for Cyber Security					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To study the technical perspective of cybercrime and cyber forensics.
- Discuss different machine learning techniques
- Understand the role of machine leaning in cyber forensics
- To study machine learning forensics

UNIT 1: Cyber Crime Introduction, Technical Perspective, Abuse & Misuse of Technologies, Cyber-Crime Challenges, Cybercrimes Classification, Tools and Methods in Cybercrime, Key Loggers and Spyware, Virus, Worms, Trojan Horses, Backdoors, DOS and DDOS Attacks, Buffer Overflow, Attack on Wireless Networks, Phishing, Cybercrime Investigation Process, Digital Forensics Investigation.	**10 Hrs.**
UNIT 2: Adversarial Machine Learning Terminology, The Importance of Adversarial ML, Security Vulnerabilities in Machine Learning Algorithms, Types of Adversarial Attacks, Attack Technique: Model Poisoning, Defense Against Poisoning Attacks, Evasion Attack, Defense Against Evasion Attacks, Mitigating Adversarial Effects, Defending against adversarial attacks	**12 Hrs.**
UNIT 3: Introduction To Machine Learning Introduction- Definition, Dataset, Training and Test Dataset, Standardization and Normalization, Machine Learning Applications in Cyber Security and Cyber Forensics, Classification and Clustering, Supervised Learning, Unsupervised Learning, Evaluation Metrics, Decision Tree, Naïve Bayes, Logistic Regression, Support Vector Machine, Random Forest, K Nearest Neighbour Classifier, K-mean Clustering, PCA, Overfitting and Underfitting.	**10 Hrs.**
UNIT 4: Applications Of Machine Learning In Cyber Forensics Machine Learning Forensics, Intrusion Detection, Anomaly Detection, Host Intrusion Detection, Network Intrusion Detection, Web Application Intrusion Detection, Approaches to Overcome Challenges in Forensics, Approaches to Prevent and Detect Cyber Security Attacks, Adversarial Attacks on Different Security Datasets.	**10 Hrs.**
	42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1- Understand cybercrime and cyber forensics.
CO2- Compare different types of cyber-attacks and how to detect them.
CO3- Compare different Machine Learning approaches.
CO4- Evaluate the performance of Machine Learning Model.
CO5- Discuss the Machine Learning Forensics.
CO6- Analyze different approaches used to detect and prevent cyber security attacks.

TEXT/REFERENCE BOOKS

1. Clarence Chio, David Freeman, "Machine Learning and Security", O'Reilly.
2. Soma Halder and Sinan Ozdemir, "Machine Learning for Cybersecurity", Packt
3. Emmanuel Tsukerman, "Machine Learning for Cybersecurity Cookbook", Packt
4. Sanjay Misra et al." Confluence of AI, Machine, and Deep Learning in Cyber Forensics", IGI Global.
5. Clarence Chio, "Machine Learning & Security: Protecting Systems with Data and Algorithms", O'Reilly.

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Program Elective Course Track 2

Industry Track

24CS432T					Agile and DevOps					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	-	-	100

COURSE OBJECTIVES

- Learn about the DevOps lifecycle and its stages, including continuous integration, continuous delivery, and continuous deployment.
- Gain practical knowledge of version control systems and configuration.
- Understand the importance of security in DevOps and learn how to integrate security practices into the CI/CD pipeline.
- Gain insights into sustainable source code management practices, to create longterm value.

UNIT 1: Understanding of Agile Process Software processes, Process Models – overview Agile Model and Various Agile methodologies - Scrum, XP, Lean, and Kanban. Scope of each model and their comparison in real-world case studies.	**10 Hrs.**
UNIT 2: Introduction to DevOps Definition and principles of DevOps, Benefits of DevOps adoption, DevOps culture and mindset, DevOps Lifecycle, Overview of the DevOps lifecycle stages, Continuous Integration (CI) and Continuous Delivery (CD), Automated testing and deployment, Version Control and Configuration Management, Introduction to version control systems (e.g., Git), Branching and merging strategies, Configuration management tools (e.g., Ansible, Puppet)	**10 Hrs.**
UNIT 3: Integration and Build Automation Introduction to Infrastructure as code (IaC) concepts and benefits, Infrastructure provisioning tools (e.g., Terraform, Cloud Formation), Managing infrastructure configurations, setting up CI/CD pipelines, Building and packaging applications, Automated testing and code quality checks, Continuous Deployment and Release Management, Strategies for releasing software updates, Managing deployment environments, Release orchestration and rollback strategies	**10 Hrs.**
UNIT 4: SCM and GIT Fundamentals and Security Overview of source code management and its role in software development, Version control systems and their benefits, Introduction to centralized and distributed SCM, Understanding repositories, commits, and revisions, Exploring the concepts of working directory, staging area, and remote repositories, Introduction to branching and tagging, Introduction to Git and its key features, Setting up a Git repository, Committing changes and viewing history, Working with remote repositories, Cloning, pushing, and pulling changes, Collaboration workflows and managing multiple contributors. Importance of monitoring in DevOps, implementing application and infrastructure monitoring, Log aggregation and analysis, Security and Compliance, DevOps security principles and practices, Implementing security controls in CI/CD pipelines, Compliance considerations in DevOps.	**12 Hrs.**
	42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1: Understand the principles and benefits of DevOps, including its role in software development and deployment, enabling them to comprehend its significance in modern software engineering practices.

CO2: Develop an understanding of compliance requirements and learn how to incorporate them effectively into DevOps processes, ensuring adherence to regulatory standards and organizational policies.

CO3: Develop a comprehensive understanding of source code management concepts, including its objectives (such as version control and collaboration) and components (such as repositories, branches, and commits), facilitating efficient and organized software development workflows.

CO4: Aain an understanding of management principles and techniques relevant to source code, including source code planning and optimization strategies, enabling them to effectively oversee and optimize the development process.

CO5: Acquire knowledge of source code management practices and their impact on the team's environment, allowing them to assess, implement, and adapt best practices for efficient collaboration, version control, and code quality assurance within the development team.

CO6: Develop proficiency in utilizing DevOps tools and technologies to automate software development, testing, and deployment processes, enabling students to streamline workflows, increase productivity, and enhance the overall efficiency of software delivery pipelines.

TEXT/REFERENCE BOOKS

1. Gene Kim, Kevin Behr, and George Spafford, "The Phoenix Project: A Novel About IT, DevOps, and Helping Your Business Win",Revolution Press.
2. Jez Humble and David Farley, “Continuous Delivery: Reliable Software Releases through Build, Test, and Deployment Automation", Addison-Wesley Professional.
3. Emma Jane, “GIT for Teams”, O'Reilly Media.

24CS437T					Web Testing					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	-	-	100

COURSE OBJECTIVES

- Understand applicable theories connecting cybersecurity to human behaviour.
- Critique the purpose and goals of vulnerability assessments and penetration testing.
- Interpret and differentiate cyber threats and exploits in a penetration testing context.
- Describe and distinguish key phases of ethical hacking: reconnaissance, scanning, gaining access, maintaining access, and covering the tracks.

UNIT 1 Principle of Security Testing Explain secure applications and the needs for testing them; model and methodologies to different applications. Different system, recognise project security threats. The following are the tenets of security testing: non-repudiation, availability, authenticity, authorization, and confidentiality. Network security, system software security, client-side application security, and server-side application security are the main areas of attention for security testing.	**10 Hrs.**
UNIT 2 Types of Security Testing Vulnerability Scanning: programme that automatically examines a system to find patterns of known vulnerabilities Security scanning reduces risks and errors by identifying network and system flaws. Security scanning, both automated and human Penetration testing involves simulating a hostile hacker's attack in order to look for any weaknesses that a hacker would use to try to access the system. Risk assessment includes security risk testing, risk classification (low, medium, and high), and risk-reduction strategies. To evaluate and enhance a product's, process', or project's security, gather and evaluate metrics using statistical techniques.	**11 Hrs.**
UNIT 3 Web application testing Network scanning, Determining different attacks on web application, applying the attacks as SQLi, XXS, Session Hijacking, etc and also developing different methodologies to secure application from such attacks and vulnerability present.	**11 Hrs.**
UNIT 4 Introduction to Mobile Security Overview of mobile security challenges and threats, Differences between mobile and traditional computing security, Legal and ethical considerations in mobile penetration testing, Understanding the architecture of mobile applications, Overview of mobile platforms (iOS, Android) and their security models, Introduction to mobile app components (UI, backend, APIs), Understanding static and dynamic analysis of mobile testing	**10 Hrs.**
	42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand the fundamental principles of and concepts of mobile and web application penetration testing.

CO2 - Identify different types of attacks of different applications.

CO3 - Apply different principals of security testing.

CO4 - Analyze the potential impact of identified vulnerabilities on the security posture of mobile and web applications.

CO5 - Evaluate the security posture of mobile and web applications using penetration testing methodologies.

CO6 - Create strategies to mitigate security risks and protect against common mobile and web application security threats.

TEXT/REFERENCE BOOKS

1. Gary McGraw, Software Security: Building Security, Addison-Wesley
2. 2. Julia H. Allen, Sean Barnum, Robert J. Ellison, Gary McGraw, and Nancy Mead, Software Security Engineering: A Guide for Project Managers, Addison-Wesley
3. Research papers and articles from relevant conferences and journals (e.g., IEEE Transactions on Information Forensics and Security, ACM Transactions on Embedded Computing Systems).
4. Online resources and tutorials on specific hardware security topics.

Program Elective Course Track 3

Speech, Vision & Text Track

24CS433T					Natural Language Processing					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	-	-	100

COURSE OBJECTIVES

- To understand the structure and basic operations of Natural Language Processing
- To understand the syntax of Natural languages for grouping local words for parsing
- To understand the concepts of linguistic rules and machine learning approaches for classification
- Apply language modelling techniques for improving application of NLP.

UNIT 1: Introduction to Natural Language Processing Building Blocks of NLP- NLP and Ambiguity, POS Tagging, Named Entity Recognition, Word Sense Disambiguation, Wordnet and Lexical Resources, Syntax – Constituency Parsing, Dependency Parsing, Distributional Semantics, Lexical Semantics	**10 Hrs.**
UNIT 2: Classical Text Processing Basic Text Processing, Language Modelling, Advanced smoothing for language modelling, Minimum Edit Distance, Spelling Correction, Models for Sequential Tagging – MaxEnt, CRF, NaiveBayes	**12 Hrs.**
UNIT 3: Deep Learning for NLP & LLM Representation Discovery: Word vectors, Word2Vec & GloVe, RNNs for Variable Length Sequences, Attention & Transformer, Neural & Pre-Trained Language Models, Advanced Pre-training for Language Models, GPT3 & Beyond: Few-Shot Learning, Prompt Learning	**10 Hrs.**
UNIT 4: NLP Applications Sentiment Analysis and Opinion Mining, Text Summarization, Sarcasm Detection, Information Retrieval, Question Answer System	**10 Hrs.**
	42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand the Natural Language in the form of Text
CO2 - Determine the Syntactical Structures of Natural Languages
CO3 - Evaluate the Language Model for Natural Languages
CO4 - Apply Information Retrieval Techniques to Build Search Engines
CO5 - Develop POS tagger, Parsers and Shallow Parser for Natural Languages
CO6 - Design Text Summarization, Sentiment Analysis, Sarcasm Detection, Spelling Correction Systems

TEXT/REFERENCE BOOKS

1. Yoav Goldberg Neural Network Methods for Natural Language Processing, Morgan and Claypool
2. Dan Jurafsky and James Martin Speech and Language Processing.
3. Christopher D. Manning and Hinrich Schütze, Foundations of Statistical Natural Language Processing.
4. James Allen, Natural Language Understanding, Benjamin/Cummings.
5. Natural Language Processing: A Paninian Perspective by Akshar Bharati, Vineet Chaitanya and Rajeev Sangal

24CS438T					Fundamental Concepts of Speech Processing					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	-	-	100

PREREQUISITE: Linear Algebra, Machine Learning

COURSE OBJECTIVES

- Describe the physiological and acoustic aspects of speech production.
- Implement various feature extraction techniques and understand their role in speech processing.
- Implement the principles of speech coding and compression.
- Apply language modelling techniques for improving speech recognition accuracy.

UNIT 1: Fundamentals of Digital Signals and Systems Basic concepts and digital representation of digital signals, Sampling, Quantization, Time Domain analysis of Speech Signals, signal segmentation and framing, Frequency Domain analysis, discrete Fourier transform (DFT) and the fast Fourier transform (FFT), digital filter design principles and techniques, Finite impulse response (FIR) filters and Infinite impulse response (IIR) filters.	**12 Hrs.**
UNIT 2: Introduction to Digital Speech Processing Overview of Speech Processing, Human Speech production, Digitization and recording of speech signal, Acoustic Phonetics and Articulatory Phonetics, Different categories speech sounds and Location of sounds in the acoustic waveform and spectrograms, Uniform Tube Modeling of Speech Production.	**10 Hrs.**
UNIT 3: Human Speech Production and Source Filter Model Speech Perception, Pitch and Formant Analysis, Mel-Frequency Cepstral Coefficients (MFCCs), Speech Feature Extraction Techniques, Segmental and Supra Segmental Features of Speech signal, Speech Prosody, Speech Prosody Modeling (Fujisaki Model).	**10 Hrs.**
UNIT 4: Speech Recognition and Applications Introduction to Speech Recognition, Hidden Markov Models (HMMs), Acoustic Modeling in Speech Recognition, Language Modeling for Speech Recognition, Decoding Techniques (Viterbi Algorithm), Text-to-Speech (TTS) Systems, Speaker Identification, Speaker Verification.	**10 Hrs.**
	42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand fundamental concepts of speech production, acoustics, and digital signal processing to represent speech signals.

CO2 - Distinguish feature extraction techniques and understand their role in speech processing applications.

CO3 - Illustrate a foundational understanding of Language Model to implement speech recognition system.

CO4 - Implement the principles of speech coding techniques for compression.

CO5 - Demonstrate speech synthesis techniques for text-to-speech systems.

CO6 - Apply speech processing technologies in real-world applications, including Automatic Speech Recognition (ASR) and Natural Language Processing (NLP).

TEXT/REFERENCE BOOKS

1. Lawrence R. Rabiner, Ronald W. Schafer, An Introduction to Digital Speech Processing: Foundations and Trends® in Signal Processing, Now publishers Inc
2. A. Nejat Ince, Digital Speech Processing: Speech Coding, Synthesis and Recognition, The Springer International Series in Engineering and Computer Science.
3. Branko Kovacevic, Milan M. Milosavljevic, Mladen Veinović, Milan Marković, Robust Digital Processing of Speech Signals, Springer International Publishing AG.

Program Elective Course Track 4

Computing Track

24CS434T					Cloud Computing					
Teaching Scheme					Examination Scheme					
					Theory			Practical		
L	T	P	C	Hrs/ Week	MS	ES	IA	LW	LE/Viva	Total Marks
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To provide an in-depth and comprehensive knowledge of Cloud Computing fundamental issues, technologies, applications and implementations.
- To expose the students to the frontier areas of Cloud Computing
- To motivate students to use various cloud computing environments in their future projects.
- To shed light on the issues and challenges in Cloud Computing

UNIT 1: Cloud Foundation And Overview Distributed Computing, Cluster computing, Grid computing. Cloud Service Models	**11 Hrs.**
UNIT 2: Virtualization And Load Balancing Virtualization concepts - Types of Virtualization, Introduction to Various Hypervisors, Moving VMs, Pros and cons of virtualization, Virtualization Technology examples. Distributed Management of Virtual Infrastructures, Scheduling, Capacity Management to meet SLA Requirements, Various load balancing techniques.	**11 Hrs.**
UNIT 3: Industrial Platforms And New Developments Study of Cloud Computing Systems like Amazon EC2 and S3, Google App Engine, and Microsoft Azure, Build Private/Hybrid Cloud using open-source tools. MapReduce and its extensions to Cloud Computing, Cloud Application Programming, and the Aneka Platform	**11 Hrs.**
UNIT 4: Advanced Topics In Cloud Computing Energy efficiency in clouds, Market-based management of clouds, Federated clouds, Security in Cloud Computing	**9 Hrs.**
	42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Explain the strengths and limitations of cloud computing and the possible applications for state-of-the-art cloud computing.

CO2 - Identify the architecture and infrastructure of cloud computing.

CO3 - Demonstrate the concept and role of virtualization in cloud computing.

CO4 - Provide the appropriate cloud computing solutions and recommendations as per the applications.

CO5 - Understand the tradeoffs in power, efficiency, and cost in cloud computing.

CO6 - Analyze authentication, confidentiality and privacy issues in cloud computing

TEXT/REFERENCE BOOKS

1. Rajkumar Buyya, James Broberg, Andrzej M Goscinski, Cloud Computing: Principles and Paradigms, Wiley publication
2. Toby Velte, Anthony Velte, Cloud Computing: A Practical Approach, McGraw-Hill Osborne Media
3. K. Chandrasekaran, Essentials of Cloud Computing
4. Recent publications for case studies

24CS439T					Quantum Computing					
Teaching Scheme					**Examination Scheme**					
L	**T**	**P**	**C**	**Hrs/Week**	**Theory**			**Practical**		**Total Marks**
					MS	**ES**	**IA**	**LW**	**LE/Viva**	
3	**0**	**0**	**3**	**3**	**25**	**50**	**25**	**-**	**-**	**100**

COURSE OBJECTIVES

- Learn the basics of quantum computing.
- Learn important applications of quantum information processing.
- Understand the various cryptography techniques and algorithms that use quantum computing.
- Know about the physical realization of a quantum computer and get basic knowledge of the development libraries for quantum computer programming.

UNIT 1: Introduction To Quantum Computing Introduction, quantum bits, multiple qubits, quantum gates, quantum circuits, no cloning theorem, linear algebra and postulates for quantum mechanics, density operator, Pauli operators, Bloch sphere representation.	**12 Hrs.**
UNIT 2: Quantum Information Processing Bell states, quantum entanglement, EPR inequality, quantum teleportation, quantum super dense coding, pure and mixed quantum states, quantum fidelity.	**12 Hrs.**
UNIT 3: Quantum Cryptography And Quantum Algorithms BB84 protocol, B91 protocol, Ekert protocol, quantum secret sharing, Quantum parallelism, Deutsch algorithm, Deutsch-Josza and Bernstein Vazirani Algorithms, Simon Problem, Grover's Search algorithm, Quantum Fourier Transform, Shor's Factorization Algorithm, Quantum noise and error correction.	**10 Hrs.**
UNIT 4: Quantum Computers: Physical Realization And Development Libraries Stern Gerlach experiment, Quantum computers using ion traps, superconducting circuits, and nuclear magnetic resonance, development libraries for quantum computer programming, quantum computing simulators like IBM Quantum experience, Qiskit and Quantum Development Kit.	**8 Hrs.**
	42 Hrs.

COURSE OUTCOMES

At the end of the course, students will be able to -

CO1 - Illustrate the potential of quantum computing.
CO2 - Understand qubits, entanglement, quantum gates, and their representation.
CO3 - Analyze the applications of quantum information processing, like teleportation and super dense coding.
CO4 - Describe quantum cryptography techniques.
CO5 - Compare quantum algorithms with their classical counterparts.
CO6 - Simulate basic quantum circuits on IBMQ.

TEXT/REFERENCE BOOKS

1. Michael A. Nielsen and Isaac L. Chuang, Quantum Computation and Quantum Information, Cambridge University Press.
2. Chris Bernhardt, Quantum Computing for Everyone, MIT Press.
3. Jack Hidary, Quantum Computing: An Applied Approach, Springer.
4. Robert S. Sutar, Dancing with Qubits: How quantum computing works and how it can change the world, Packt Publishing Ltd..
5. Vishal Sahni, Quantum Computing, TATA McGrawHill.

Program Elective Course Track 5

Communication Track

24CS435T					Internet of Things					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To impart knowledge about components of IoT Architecture and platforms of IoT.
- To study different types of protocols used in IoT.
- To develop real life IoT based solutions.

UNIT 1: Introduction To Internet Of Things IoT Definition, IoT characteristics, M2M and IoT, End to End IoT Architecture, Physical design of IoT, Logical design of IoT, Challenges for IoT, Interdependencies of IoT and cloud computing, Web of things, IoT data management and compute stack	**10 Hrs.**
UNIT 2: Iot PROTOCOLS Overview of IoT protocols, IoT access technologies: IEEE 802.15.4, ZigBee, LoRaWAN, IP as IoT network layer: 6LoWPAN, 6Lo, RPL, Transport layer protocols, Application layer protocols-CoAP, MQTT.	**12 Hrs.**
UNIT 3: Embedded Iot Devices Sensors and actuators for IoT applications, IoT components, Smart objects, Connecting smart objects: Communication criteria, Basics of sensor networks, Types and configurations of gateways, Specifications of IoT gateways, Digital Twin.	**10 Hrs.**
UNIT 4: Application Building With IOT Security and IoT, Data Analytics for IoT, Big Data analytics tools and technology, Various applications of IoT: Smart city, Smart grid, Smart transportation, Smart manufacturing, Smart healthcare	**10 Hrs.**
	42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1- Understand the Internet of Things and its components.
CO2- Correlate the IoT architecture with various technological resources/ enabling technologies.
CO3- Compare different protocols/communication technologies of the IoT protocol stack.
CO4- Determine I/O device interfaces, sensors, actuators & communication modules for IoT application
CO5- Analyze different security aspects for the Internet of Things.
CO6- Classify IoT applications and design IoT solutions to lead the smart world.

TEXT/REFERENCE BOOKS

1. D. Hanes, G. Salgueiro, P. Grossetete, R. Barton, J. Henry. “IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things”, Cisco Press.
2. A. Bahga, V. Madisetti. “Internet of Things: A Hands-on Approach”, Universities Press (India) Pvt. Ltd.
3. S. Vasudevan, A. Nagrajan and R.M.D. Sundaram. “Internet of Things”, Wiley India
4. Y. Kanetkar, S. Korde. “21 IoT Experiments”, BPB

24CS440T					Autonomous System					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/ Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	-	-	100

PREREQUISITE: Internet of Thing, Computer Programming, Artificial Intelligence

COURSE OBJECTIVES

- Learn the basics of autonomous systems and their technologies.
- Apply algorithms to develop autonomous system solutions.
- Integrate data and intelligence to improve traditional practices in autonomous system
- Design intelligent machines for autonomous system applications.

UNIT 1: Introduction to Autonomous Systems Introduction and Evolution of Robotics and Autonomous Systems, Basics of Robotics and Control Systems, Sensors and Actuators in Autonomous Systems, Basics of Navigation and Path Planning, Introduction to Communication Protocols for Autonomous Systems, Introduction to UAVs: Types, Applications, and Challenges, Expert System and Deep Learning in Autonomous Systems	**10 Hrs.**
UNIT 2: Perception and Sensing Overview of Perception in Autonomous Systems, IOT for Autonomous system, Sensor Fusion Techniques, Computer Vision for Object Detection and Recognition, Deep Learning for Perception in Autonomous Systems, UAV Perception Systems: Cameras, Lidar, and Radar, Case Studies and Applications in UAV Perception.	**10 Hrs.**
UNIT 3: Cross-Domain Integration of Autonomous Systems Edge and Fog Computing for Autonomous Systems, Communication Protocols and Networking for Multi-agent Systems, Human-Robot Interaction and Collaboration, Autonomous Systems in Challenging Environments (e.g., Space, Underwater), Advanced Deep Learning Techniques for Autonomous Systems: Recurrent Neural Networks (RNNs), Generative Adversarial Networks (GANs), etc.	**12 Hrs.**
UNIT 4: Case Studies and Application of Autonomous System Case studies in Transportation: Autonomous Vehicles in Urban and Rural Environments, Healthcare: Robotics and AI in Medical Diagnosis and Treatment, Agriculture: Drones and Robotics for Precision Farming, Environmental Monitoring: UAVs and Robotics for Surveillance and Conservation, Smart Cities: Integration of Autonomous Systems for Urban Development.	**10 Hrs.**
	42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Gain an understanding of the foundational components and key technologies underpinning autonomous systems, including robotics, UAVs, navigation, communication, fog/edge computing, and data analysis.

CO2 - Apply a variety of computational techniques and algorithms to design, develop, and deploy autonomous systems, ensuring proficiency in tasks such as path planning, perception, decision-making, and control.

CO3 - Integrate data from sensors, actuators, and other sources to enable intelligent decision-making and enhance operational efficiency in diverse applications of autonomous systems, thereby transforming traditional business models and processes.

CO4 - Develop intelligent and automated machines for traditional industries, leveraging advancements in robotics, AI, and automation to optimize production processes, improve quality, and reduce human intervention.

CO5 - Utilize data analytics and artificial intelligence to create Smart World solutions, enabling autonomous systems to enhance urban infrastructure, transportation networks, healthcare delivery, environmental monitoring, and other aspects of daily life.

CO6 - Understand the concept, significance, and strategies for achieving sustainable development in the context of autonomous systems, considering ethical, social, and environmental implications to ensure responsible deployment and utilization of these technologies.

TEXT/REFERENCEBOOKS

1. "New autonomous systems." by Cardon, Alain, and Mhamed Itmi. by John Wiley & Sons.
2. "The The Complete Edition–Software Engineering for Real-Time Systems: A software engineering perspective toward designing real-time systems" by Cooling, Jim. Packt Publishing Ltd.
3. "The Autonomous System: A foundational synthesis of the sciences of the mind." by De Gyurky, Szabolcs Michael, and Mark A. Tarbell. John Wiley & Sons.